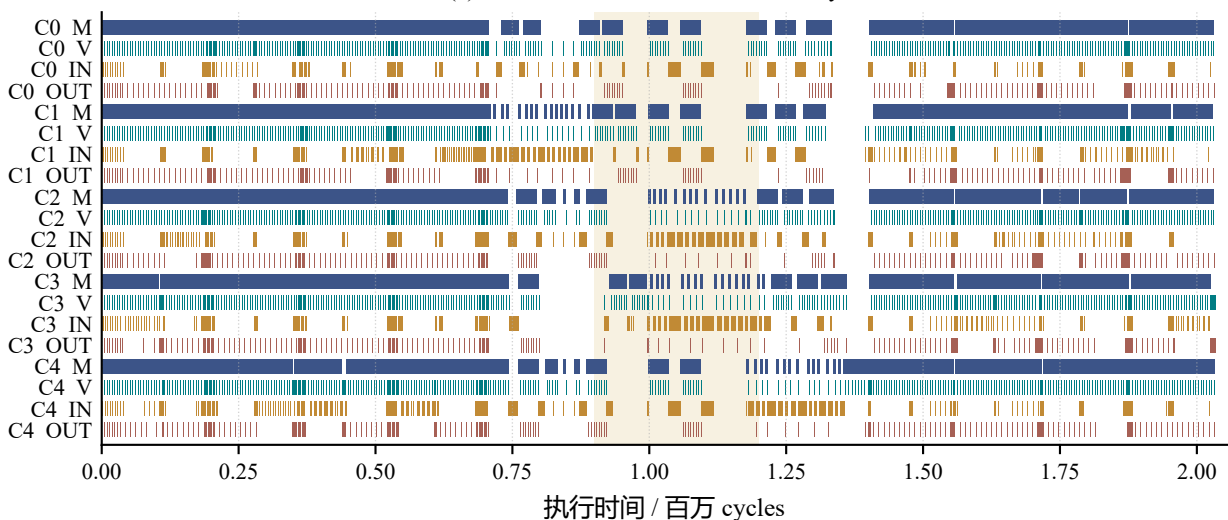
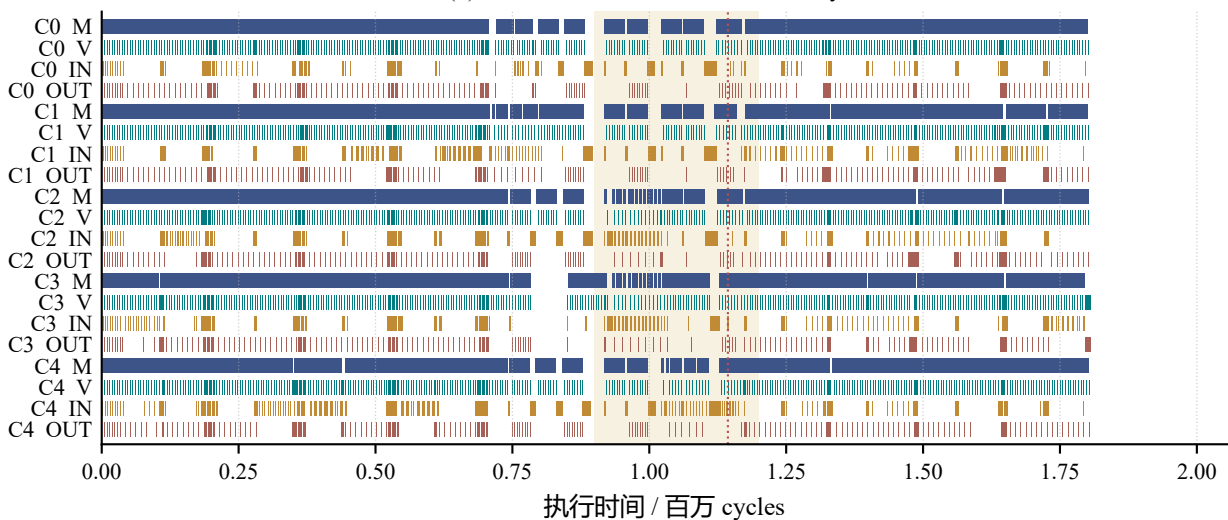


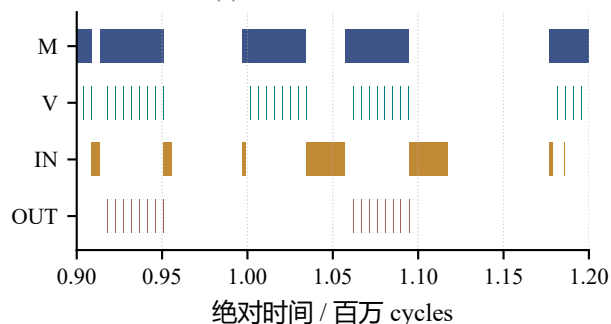
(a) 无L2: 固定PB; T = 2,034,983 cycles



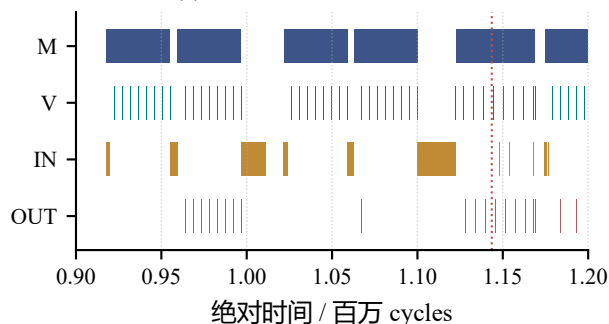
(b) 有L2: 同一PB; T = 1,806,253 cycles



(c) 无L2 / 核心0 / 局部



(d) L2 / 核心0; 红线e2428见F27



阴影为共同绝对时间窗; 红线是L2全局缓存事件, 不指定某Pipe的因果关系。